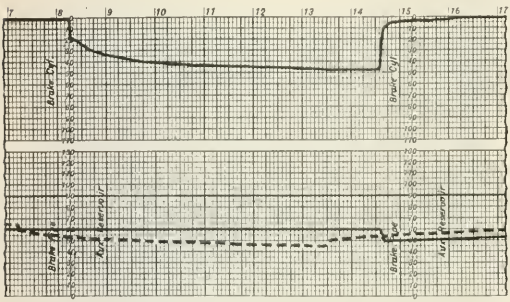
Automatic Straight Air Brake System

Description of the Design and Operation of a New Triple Valve for Passenger and Freight Equipment

A NEW air brake system has recently been perfected by the Automatic Straight Air Brake Company, 14 Wall street, New York, for freight and passenger equipment, which contains many new and interesting features of operation. The purposes of this brake are to give rapid serial action to the brakes throughout a train, to maintain a constant and uniform brake cylinder pressure regardless of piston travel, to permit a variation of brake cylinder pressure at the will of the engineman, to provide a proper and quick release of the brakes for any brake pipe reduction, to



Service Application of the Brakes with Quick Release

The top curve represents brake cylinder pressure, the full line in the lower set of curves represents the pressure in the auxiliary reservoir, while the dash line shows the pressure in the brake pipe. The pen indicating the brake pipe pressure is set a distance of one minute back of the pens indicating the brake cylinder and auxiliary reservoir pressures. The above curves show that as the brake pipe pressure is reduced, the brake cylinder pressure of 20 lb. is obtained at once, which is increased on further reduction of the brake pipe pressure to about 50 lb., the pressure in the auxiliary reservoir remaining constant. With the building up of the brake pipe pressure, a small amount of air from the auxiliary reservoir is released to the brake pipe to hasten the release, the rapidity of release being indicated by the brake cylinder curve.

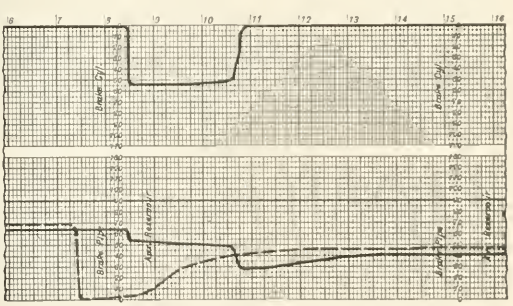
provide for a full emergency application of the brakes at any time, to provide a graduated or quick release as desired, and to provide an economical use of air.

This brake system provides a quick action passenger brake, one brake cylinder being used for a service application and two brake cylinders for an emergency application of the brakes. As the triple valve is capable of compensating for varying volumes in brake cylinders, a second brake cylinder can be added to existing freight equipment for empty and load braking, the braking system retaining at the same time all of its functions and principles.

The brake is operated by the engineman in the same manner as is common with present day practice. The equipment can be used interchangeably with other existing equipment. The main features of this new brake are found in an entirely new triple valve with its auxiliary, service and quick action reservoirs. The auxiliary reservoir is of the same volume as that ordinarily used, the service reservoir has a volume of 2,100 cu. in. and the quick action reservoir a volume of 200 cu. in., which with an additional volume due to brake pipe connections, gives an increase in volume of about 2,300 cu. in. per car on 10-in. freight equipment. The service reservoir is used for service and emergency applications of the brake, the auxiliary reservoir is used for an emergency application of the brake and for a quick release of the brake in contrast to a graduated release. The quick action reservoir is used only in making an emergency application of the brakes. The triple valve is made up of disk valves and diaphragms, no slide valves or pistons being used.

Due to its construction and operation this brake has the characteristics of a straight air brake and at the same time is automatically operated. The straight air features are obtained through the fact that with every application of the brakes air is exhausted from the brake pipe under each car. In case of a service application air from the brake pipe is exhausted into the brake cylinder and in case of an emergency application the air from the brake pipe is exhausted to the atmosphere. In the first case the air is not wasted but is used to build up the pressure in the brake cylinder in conjunction with the supply of air from the service reservoir. In both cases the rapidity of serial action is increased due to the fact that the brake pipe pressure is reduced locally at each car.

The design of the triple valve is such that when fully charged the pressure in the brake pipe acting on the underside of a diaphragm balances the pressure in the auxiliary reservoir acting on the upper side. A reduction in brake pipe pressure causes the auxiliary reservoir pressure to force the diaphragm downward admitting air to the brake cylinder from the brake pipe and service reservoir. The air in the brake cylinder acts on a second diaphragm which is connected to the first and which is of one-half its area. The pressure in the auxiliary reservoir remains unbalanced forcing the diaphragm down until the force exerted by the brake pipe pressure on the underside of this diaphragm plus force exerted by the air in the brake cylinder on its diaphragm exceeds it. The diaphragm will then be raised and the supply of air to the brake cylinder cut off. Thus it will be seen



Emergency Application of the Brakes

In this case the brake pipe pressure was reduced to zero, giving an instantaneous brake cylinder pressure of 55 lb. with an accompanying decrease in auxiliary reservoir pressure. At release more air is taken from the auxiliary reservoir to aid the release as described under the quick release operations, the brake being released in about 10 seconds. Due to the construction of the triple valve, the auxiliary reservoir pressure is then raised, through the charging port, with the brake pipe pressure, the difference in the amount being due to the spring used on the auxiliary reservoir diaphragm. This record was obtained from car 61 in the 100-car test rack described in the text.

that the brake cylinder pressure bears a direct relation to the brake pipe reduction and is not affected by the brake piston travel or brake cylinder leakage. By regulating the brake pipe pressure, any brake cylinder pressure may be obtained.

Each triple valve is provided with means for making a graduated or quick release. The graduated release is obtained by building up the brake pipe pressure from the locomotive. The quick release feature is obtained by raising the